Ranjan Khadka

Academic Curriculum Vitae

Crookston, Minnesota | khadk046@crk.umn.edu | (218) 275-0287

GitHub | Portfolio | LinkedIn

EDUCATION

University of Minnesota Crookston **Expected May 2027**

Bachelor of Science in Software Engineering

Minors: Cybersecurity and Information Assurance; Information Technology Management

Cumulative GPA: 3.968/4.000

Selected completed coursework: Artificial Intelligence; Database Management Systems; Cybercrime and Digital Forensics; Secure Data Collection and Retrieval; Discrete Mathematics; Calculus I.

In progress Fall 2026: Data Structures and Algorithms; Software Design and Architecture; Undergraduate Research; Senior Project I; Introduction to Statistics.

RESEARCH EXPERIENCE

Object Detection and Classification Under Adverse Weather and Low-Light Conditions

Undergraduate Researcher, University of Minnesota Crookston | Fall 2026 - Present

Supervisors: M M Mahabubur Rahman and Md Nurul Amin

- Investigating object detection and classification in hazy, snowy, and low-light conditions representative of the Upper Midwest, especially Minnesota and North Dakota.
- Developing the research toward an undergraduate paper; work is ongoing.

Illegal Parking Detection in Aerial Video **Summer 2026**

Research project and unpublished manuscript

Supervisor: Md Nurul Amin

- Developed a prototype combining YOLO11n vehicle detection, ByteTrack tracking, stationary-state analysis, and polygon-based parking rules.
- Fine-tuned YOLO11n on 123 aerial training images using CPU-only hardware; obtained vehicle-detection mAP@0.5 of 0.757 on 31 validation images.
- Compared alert logic with a MOG2 baseline in two controlled demonstrations; documented limitations from missing real-world positive-event labels.

TEACHING AND ACADEMIC SUPPORT

Mathematics Teacher **2022 - 2024**

Suryodaya Secondary School

- Taught mathematics to middle school students while studying at Trichandra Multiple Campus.

Chemistry and Physics Laboratory Support (Informal)

Trichandra Multiple Campus

- Provided informal support in the chemistry and physics laboratories while attending the campus.

Individual Tutor

- Provided one-to-one academic support to a Grade 9 student from a marginalized family, helping her improve her grades.

HONORS AND AWARDS

Chancellor's List: Spring 2024, Spring 2025; Dean's List: Fall 2024, Spring 2026

First Place, Huskies Hack: IdeaForge, St. Cloud State University | March 2026

Second Place, Southwest MN Hacks: Fall 2026 | Alignt | September 2026

Facilities Student Employee of the Year | University of Minnesota Crookston | 2025

Inclusion, Belonging, and Excellence Award | University of Minnesota Crookston | 2024

TECHNICAL EXPERIENCE

Software Engineering Intern **May 2025 - Present**

University of Minnesota Crookston Sustainability Department

- Collaborated in a two-person team to develop and deploy UMC-Green, a campus marketplace for donating and receiving reusable items.
- Migrated MongoDB to SQL, integrated backend services, APIs, and authentication, and configured Red Hat Enterprise Linux and Nginx deployments; worked with Railway and Vercel hosting.

SELECTED SOFTWARE PROJECTS

ARIA Adaptive Responsive Intelligent Assistant **Fall 2026 - Present**

Senior capstone design | Backend Development and LLM Infrastructure

- Collaborating on requirements and system design for a voice- and text-enabled desktop assistant for productivity tasks.
- Planning a Python backend and LLM infrastructure with a self-hosted language-model service and a hosted-model fallback.
- Design covers command orchestration, Google Calendar and Gmail integration, OAuth 2.0 authorization, local SQL storage, and confirmation for sensitive actions.

Alignt **September 2026**

Workforce planning prototype | React, Node.js, AI API integrations

- Developed backend integrations for a workforce-planning prototype combining a React dashboard with AI-assisted role and skill analysis.
- Worked on application deployment and CI/CD workflows for the hackathon prototype.

github.com/RanjanKhadka012/Alignt

11Bunkers **March 2026**

Hackathon team project | React, Node.js, Claude API, ElevenLabs API

- Built the Node.js and Express backend for an interactive storytelling and social survival game.
- Integrated Claude-generated narration and ElevenLabs speech synthesis into the game.

github.com/RanjanKhadka012/Eleven_Bunkers

TECHNICAL SKILLS

Programming: Python, Java, C, C++, JavaScript, SQL, HTML, CSS

Web and databases: React, Node.js, Express.js, REST APIs, MySQL, MongoDB

Systems and tools: Linux, Red Hat Enterprise Linux, Git, Docker, Nginx, Railway, Vercel, CI/CD

Data processing: Hadoop, Spark, Spark SQL, HDFS

Computer vision: YOLO11n fine-tuning, ByteTrack, MOG2 background subtraction, detection evaluation

ADDITIONAL EXPERIENCE

IT Help Desk Technician **October 2024 - Present**

University of Minnesota Crookston

- Diagnose and resolve hardware, software, networking, account access, and device configuration issues; install applications and document support tickets.

Community Advisor **August 2026 - Present**

Residential Life, University of Minnesota Crookston

- Mentor residential students, connect them with campus resources, and plan activities that support community participation.
- Respond to student concerns, conflicts, and emergencies while following university procedures.

Student Supervisor and Assistant Manager **Summers 2024 - 2026**

Facilities and Grounds, University of Minnesota Crookston

- Supervised student employees, assigned work, and supported scheduling, equipment management, and safe campus operations.

LEADERSHIP AND SERVICE

Founding Member, Innovation and Technology Alliance **2024 - Present**

Vice Student Chair, IEEE Club **2024 - Present**

New International Student Orientation Volunteer, University of Minnesota Crookston

Organizing Committee Member, Model United Nations **2021**

Delegate, Model United Nations **2019**

Class Representative **Grades 11-12**

School Captain **Grades 9-10**

COMMUNITY ENGAGEMENT

Co-Founder and Organizer, Bhetum

- Co-organized interdisciplinary youth meetups and talks with guest speakers from digital marketing, chess leadership, and content creation.
- Continued community hikes and litter collection around Kathmandu to promote environmental awareness.

Hiking and Cleanup Volunteer, Walnepalwalk

- Participated in hikes around the hills of Kathmandu, collecting litter along the routes to raise environmental awareness.

School Library Donation and Expansion Volunteer

- Helped a local school receive a book donation from an Australian university.
- Supported the school library's expansion by categorizing and organizing the donated books.

ATHLETIC ACHIEVEMENTS

4th International Karate Championship **May 10-12, 2018**

Represented Nepal in the open-club international karate competition at the Table Tennis Indoor Hall, Lainchour, Kathmandu, Nepal.

District-Level Karate Champion **2017, 2018**

- Represented school in karate competitions at multiple levels.